

AIMS-TBI - Automated Identification of Moderate-Severe Traumatic Brain Injury Lesions: Structured description of the challenge design

Remark: This challenge has been slightly modified. All changes are highlighted in blue.

CHALLENGE ORGANIZATION

Title

Use the title to convey the essential information on the challenge mission.

AIMS-TBI - Automated Identification of Moderate-Severe Traumatic Brain Injury Lesions

Challenge acronym

Preferable, provide a short acronym of the challenge (if any).

AIMS-TBI

Challenge abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Moderate to Severe Traumatic Brain Injury (msTBI) is caused by external forces (eg: traffic accidents, falls, sports) causing the brain to move rapidly within the skull, resulting in complex pathophysiological changes. Primary injuries arising from the initial forces (e.g., haematoma, hemorrhages, and contusions) (Mckee & Daneshvar, 2015), subsequently induce a cascade of secondary injuries (e.g., gliosis, encephalomalacia, (Maas et al., 2008)) that can include life threatening disorders such as raised intracranial pressure, requiring acute surgical intervention (Bullock et al., 2006). Each of these primary, secondary and surgery related processes can cause structural deformation in the brain. Each patient with msTBI has a unique accumulation of these structural changes, contributing to extremely heterogeneous lesions, considered a hallmark of msTBI (Covington & Duff, 2021). These lesions differ from other common brain pathologies (stroke, multiple sclerosis (MS), brain tumor) as they can be both focal or diffuse, varying in size, number and laterality, extending through multiple tissue types (e.g., grey matter (GM), white matter (WM), cerebrospinal fluid (CSF)), and can also occur in homologous regions of both hemispheres. Lesions such as these can complicate image registration, normalization, and are known to introduce both local and global errors in brain parcellation (Diamond et al., 2020; King et al., 2020). While several tools exist to compensate for lesions in neuroimaging pre-processing (i.e., the high definition brain extraction tool (HD_Bet) (Isensee et al., 2019), virtual brain grafting (VBG)(Radwan et al., 2021)), most require time consuming manual creation of lesion masks and subsequent manual quality assessment. Furthermore, in our experience, automated methods that have been developed for lesions of different etiologies (e.g. stroke, tumors [Sanjuán et al., 2013]) often underperform in TBI cases.

In the absence of appropriate processing tools, current lesion compensation techniques in msTBI include ignoring

the lesions (resulting in unreliable findings), excluding msTBI patients with large lesions (limiting the generalizability), or manual segmentation of lesions prior to analysis (time consuming). Additionally, manual segmentation is often only feasible in smaller, single-site studies which lack the statistical power to perform subgroup analyses (i.e., divide the TBI sample according to lesion characteristics). These restrictive approaches limit the ability to investigate how factors such as type of injury (axonal injury, focal lesions, and diffuse microlesions), severity (mild, moderate, severe), and presence of comorbid injuries or complications (especially those that affect pulmonary and cardiovascular function, see (Crawford et al., 2019) and post-traumatic seizure [Bennett et al., 2017; Liesemer et al., 2011]) impact patient's functional outcomes. To capture this information, msTBI researchers need access to an accurate, automatic lesion segmentation algorithm trained on large multicohort MRI datasets, where images are aggregated across sites. Whilst a handful of TBI specific algorithms exist, they require either multiple image types (Kamnitsas et al., 2017) (T1, T2, FLAIR, GE & PD) or can run on only CT images (Jain et al., 2019). The necessity for multiple image types limits the applicability of such tools to large-scale consortia aggregating common MRI scans across sites.

The AIMS-TBI challenge leverages data shared within the Enhancing NeuroImaging Genetics through Meta-Analysis (ENIGMA) Consortium (Thompson et al., 2022) Traumatic Brain Injury working group, specifically, the Pediatric msTBI and Adult msTBI subgroups. Therefore, the AIMS-TBI challenge will focus on identifying lesions in T1-weighted (T1w) MRI data only as it is both the most common MRI scan across our ENIGMA TBI consortium, and exhibits less parameter variation (e.g. 1 mm³ voxel size was relatively common in our previous published work, Dennis et al., 2023; Keleher et al., 2022) compared to other MRI modalities (e.g., diffusion MRI) which therefore results in more comparable data collected across sites.

The inaugural AIMS-TBI Challenge was held at the 2024 MICCAI conference and it was held again in 2025. The final AIMS-TBI 2025 dataset comprised a total of 892 images, including 553 images for training, 100 for validation, and 239 for testing. In the validation phase of the challenge, 15 teams uploaded 60 entries, with 9 teams submitting final models in the test stage. The best performing model achieved an average Dice score of 0.637 overall, and the highest Dice for the lesion-only files being 0.515. This result represented an improvement over the first year, yet also highlighted substantial room for continued development. This year we will separate the tasks of segmentation and detection, identifying which subjects are lesion-free in the training data and focusing the performance metrics on the cases with lesions and having dual leaderboards for each task. Additionally, this year we will make multimodal MRI data available for teams to use in training their algorithms.

Challenge keywords

List the primary keywords that characterize the challenge.

traumatic brain injury, lesion, MRI, segmentation, TBI

Year

2026

Novelty of the challenge

Briefly describe the novelty of the challenge.

This would be the 3rd year AIMS-TBI is active, and it fills a need in the field as currently available automated lesion segmentation algorithms (for TBI or other brain lesions) have poor performance for TBI. The novelty this year is

that we will have multimodal MRI available during training. T1-weighted MRI remains the primary modality available for all cases. For some cases, any combination, or all of diffusion MRI, T2-weighted MRI, FLAIR, and SWI will also be available. Participants will have the option to train on the additional modalities but will only be tested on detection and segmentation performance on T1-weighted MRI. Additionally, this year we will have dual leaderboards, one for detection and one for segmentation, as this was a major theme in the discussion among participants during AIMS-TBI 2025.

Task description and application scenarios

Briefly describe the application scenarios for the tasks in the challenge.

The tasks are to detect and segment lesions due to traumatic brain injury (TBI) from brain MRI. Application scenarios include:

- Research: the presence of lesions complicates neuroimaging processing, limiting the methods that could be used and limiting the generalizability of findings
- Intervention/treatment planning: accurate segmentation of lesions currently requires expert input, which is sometimes limited outside of academic medical centers. Automated algorithms could assist with planning and tracking interventions or treatments.

FURTHER INFORMATION FOR CONFERENCE ORGANIZERS

Workshop

If the challenge is part of a workshop, please indicate the workshop.

This challenge is associated with the full day BraTS challenges and the BrainWorks workshop. The 2 hour duration includes a shared keynote lecture with the BrainWorks workshop.

Duration

How long does the challenge take?

2 Hours

In case you selected half or full day, please explain why you need a long slot for your challenge.

N/A

Expected number of participants

Please explain the basis of your estimate (e.g. numbers from previous challenges) and/or provide a list of potential participants and indicate if they have already confirmed their willingness to contribute.

Based on the number of teams who have participated in related challenges (BraTS - 30, ATLAS - 14, ISLES - 110, AIMS-TBI 2025 - 15), we estimate that at least 15 teams will participate.

Publication and future plans

Please indicate if you plan to coordinate a publication of the challenge results.

Participating teams will be asked to submit a 8-12 page paper on their algorithms and results after the validation phase, due July 31st, as part of the challenge. In addition to the publications submitted as part of the challenge, we will coordinate a publication after the conclusion of the challenge with members from the 3 top-performing teams.

MICCAI LNCS proceedings

Indicate if you want to offer MICCAI Springer LNCS proceedings to the participants. Publishing a proceedings volume is optional and at the discretion of each challenge's organizers. At a minimum, organizers must ensure that a description of each participant's submission is publicly available. Organizers who wish to publish MICCAI Springer LNCS proceedings must adhere to the MICCAI Satellite events publication process.

Yes

Collaboration with European Society of Radiology (ESR)

In collaboration with European Society of Radiology (ESR), we announce special clinical interest topics with associated clinicians who can help with the preparation of the proposals; the best 3 challenge proposals on these topics will get the opportunity to present their challenges at the European Congress of Radiology (ECR) 2027 in a special session. If you want to organize a challenge in collaboration with ESR on one of these topics, please reach out to the MICCAI Challenges Team (miccai-challenges-2026@dkfz-heidelberg.de) and we will put you in contact with the corresponding clinician.

Challenge in collaboration with ESR. Ticking 'Yes' implies that the challenge has been prepared in collaboration with the clinical contact point.

No

Space and hardware requirements

Organizers of on-site challenges must provide a fair computing environment for all participants. For instance, algorithms should run on the same computing platform provided to all.

Our challenge will be hosted on the Synapse platform. All data will be available through the Synapse and Box, and we will use the platform for validation and testing. Projectors will be needed during the 2 hour challenge, and possibly microphones/speakers depending on the size of the room.

TASK 1: Detection

SUMMARY

Abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Moderate to Severe Traumatic Brain Injury (msTBI) is caused by external forces (eg: traffic accidents, falls, sports) causing the brain to move rapidly within the skull, resulting in complex pathophysiological changes. Primary injuries arising from the initial forces (e.g., haematoma, hemorrhages, and contusions) (Mckee & Daneshvar, 2015), subsequently induce a cascade of secondary injuries (e.g., gliosis, encephalomalacia, (Maas et al., 2008)) that can include life threatening disorders such as raised intracranial pressure, requiring acute surgical intervention (Bullock et al., 2006). Each of these primary, secondary and surgery related processes can cause structural deformation in the brain. Each patient with msTBI has a unique accumulation of these structural changes, contributing to extremely heterogeneous lesions, considered a hallmark of msTBI (Covington & Duff, 2021). These lesions differ from other common brain pathologies (stroke, multiple sclerosis (MS), brain tumor) as they can be both focal or diffuse, varying in size, number and laterality, extending through multiple tissue types (e.g., grey matter (GM), white matter (WM), cerebrospinal fluid (CSF)), and can also occur in homologous regions of both hemispheres. Lesions such as these can complicate image registration, normalization, and are known to introduce both local and global errors in brain parcellation (Diamond et al., 2020; King et al., 2020). While several tools exist to compensate for lesions in neuroimaging pre-processing (i.e., the high definition brain extraction tool (HD_Bet) (Isensee et al., 2019), virtual brain grafting (VBG)(Radwan et al., 2021)), most require time consuming manual creation of lesion masks and subsequent manual quality assessment. Furthermore, in our experience, automated methods that have been developed for lesions of different etiologies (e.g. stroke, tumors [Sanjuán et al., 2013]) often underperform in TBI cases.

In the absence of appropriate processing tools, current lesion compensation techniques in msTBI include ignoring the lesions (resulting in unreliable findings), excluding msTBI patients with large lesions (limiting the generalizability), or manual segmentation of lesions prior to analysis (time consuming). Additionally, manual segmentation is often only feasible in smaller, single-site studies which lack the statistical power to perform subgroup analyses (i.e., divide the TBI sample according to lesion characteristics). These restrictive approaches limit the ability to investigate how factors such as type of injury (axonal injury, focal lesions, and diffuse microlesions), severity (mild, moderate, severe), and presence of comorbid injuries or complications (especially those that affect pulmonary and cardiovascular function, see (Crawford et al., 2019) and post-traumatic seizure [Bennett et al., 2017; Liesemer et al., 2011]) impact patient's functional outcomes. To capture this information, msTBI researchers need access to an accurate, automatic lesion segmentation algorithm trained on large multicohort MRI datasets, where images are aggregated across sites. Whilst a handful of TBI specific algorithms exist, they require either multiple image types (Kamnitsas et al., 2017) (T1, T2, FLAIR, GE & PD) or can run on only CT images (Jain et al., 2019). The necessity for multiple image types limits the applicability of such tools to large-scale consortia aggregating common MRI scans across sites.

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Meta-Analysis (ENIGMA) Consortium (Thompson et al., 2022) Traumatic Brain Injury working group, specifically, the Pediatric mTBI and Adult mTBI subgroups. Therefore, the AIMS-TBI challenge will focus on identifying lesions in T1-weighted (T1w) MRI data only as it is both the most common MRI scan across our ENIGMA TBI consortium, and exhibits less parameter variation (e.g. 1 mm³ voxel size was relatively common in our previous published work, Dennis et al., 2023; Keleher et al., 2022) compared to other MRI modalities (e.g., diffusion MRI) which therefore results in more comparable data collected across sites.

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Keywords

List the primary keywords that characterize the task.

T1-weighted MRI, TBI, traumatic brain injury, lesion, detection

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Emily Dennis - University of Utah

Evelyn Deutscher - Deakin University

Elisabeth Wilde - University of Utah

Matthew Pease - Indiana University

Ujjwal Baid - Emory University

Spyridon Bakas - Indiana University

Adrian Onicas - University of Utah

b) Provide information on the primary contact person.

Emily Dennis, emily.dennis@hsc.utah.edu

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Yes Matthew Pease is a neurosurgeon with experience in surgical interventions for TBI. He assisted with designing the challenge in 2025 and gave a lecture during the challenge on lesions in TBI from a clinical perspective. He continues to play a central role in designing the challenge to be clinically relevant and biologically based, and assists with the review of the reference standard and interpretation of results.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place. Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

Repeated event with annual fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

MICCAI 2026

b) Report the platform used to run the challenge.

Synapse

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

New challenge website has not been established yet. We are shifting from Grand Challenge (which we used in 2024 and 2025) to Synapse as the challenge is run in closer coordination with BraTS

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

Users are allowed to conduct preprocessing steps if they are done with a readily available tool.

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

Publicly available data is allowed.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard

d) Define the award policy. In particular, provide details with respect to challenge prizes.

The top 3 performing teams will receive a cash prize - \$500 USD for first place, \$325 for second place, \$175 for third place.

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

Algorithm performance will be posted on the challenge website during the validation phase and updated with each submission. Final performance will be announced at the MICCAI in-person challenge event and the top 3 teams will be listed on the challenge website at this time.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

Participating teams will be asked to submit a 8-12 page paper on their algorithms and results after the validation phase, due July 31st, as part of the challenge. The top three teams will be invited to work with the challenge organizers on a challenge paper to be submitted to a medical imaging journal after the conclusion of the challenge.

Teams may publish any additional papers after submission of the challenge paper.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Algorithm container on the Synapse platform. <https://www.synapse.org/aimstbi2026> - (Website will be publicly visible after the challenge approval).

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

We plan to release a validation subset 6 weeks after the release of the training data. The ground truth data for the validation set will not be released, but 3 submissions will be allowed for groups to fine tune their algorithms prior to the final test.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

16 March 2026 (or earlier): Registration opens

Release of Training data

Participants will be able to register for the challenge in synapse.org, from the date of its potential acceptance (either Feb 9, or Mar 16, 2026) until the short paper submission deadline (July 31, 2026).

30 April 2026: Release of Validation data

31 July 2026: Submission deadline for Paper and Containerized method

Paper should include motivation, literature, method description, and results on training and validation data. The containerized method will be evaluated on the hidden testing data by the organizers, only for participants with submitted short papers. Ranking of all participating methods, following statistical significance assessment based on multiple permutation testing.

7 August 2026: Invitations are sent

Inviting participants with a valid submitted paper and an evaluated container to present at the conference. Teams will be notified at this point of their rank and the top 3 teams will be invited to give an oral presentation.

14 August 2026: Paper reviews are available

19 August 2026: Release of summary testing results to each participant.

Participants will receive summary testing data results, in order to report those in their papers.

21 August 2026: Camera-ready paper and Copyright form submission deadline

1 September 2026: Organizers to send all proceedings material to the MICCAI 2026 Satellite Event committee for inclusion in the MICCAI LNCS Springer Proceedings.

4-8 October 2026: Challenge at MICCAI

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

University of Utah IRB has acknowledged the ENIGMA projects as not human subjects due to data de-identification. The University of Utah has DUAs/MTAs/CRADAs/SCCs in place with each site that has contributed data. Data will be further anonymized with the following steps:

- MRI data are defaced using pydeface with any subject or site data stripped from the metadata
- New ID codes were created, with original subject IDs recoded with the key securely held at the University of Utah
- Data is pooled across sites with no way to re-identify site
- The only demographic data provided along with the MRI data are age (rounded to the nearest year), sex, and time since injury (in weeks)

With these steps, we are confident that subject/site re-identification will be impossible. Participating challenge teams will need to complete a MOU with the University of Utah prior to receiving access to the data.

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)
- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

A previous script for evaluating results can be found here: <https://github.com/adionicas/AIMS-TBI-challenge>. A more formal, updated script will be released through our Synapse website.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

The participants are required to submit their containerized algorithm, during or after the validation phase. Specific instructions for the containerization will be provided after the challenge approval. All challenge participants will be required to accept an agreement through the synapse.org website that participation in the testing phase of the challenge, will automatically mean that we can make their containerized method publicly available through our challenge webpage and a collective toolkit we are putting together.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

The National Institutes of Health funds this work. Emily Dennis, Evelyn Deutscher, and Adrian Onicas will have access to the test case labels.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Research,Treatment planning,Diagnosis,Assistance,Surgery,Intervention planning,Education,Training

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation

- Tracking

Detection

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort would be patients who have sustained a moderate-severe TBI who have MRI data.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The challenge cohort includes subjects from the ENIGMA Pediatric mTBI and ENIGMA Adult mTBI working groups who have MRI data that has been shared with the University of Utah.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

T1-weighted brain MRI, diffusion MRI, FLAIR, SWI, T2

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

Scan manufacturer and segmentation mask.

b) ... to the patient in general (e.g. sex, medical history).

Patient age, sex, time since injury

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

Brain MRI scan

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

3D lesions in the brain

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Find highly accurate lesion detection algorithm for lesions due to mod/sev TBI

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

The MRI data used in this challenge come from multiple MRI scanner manufacturers (GE, Siemens, Philips), multiple scan models, and differing sets of parameters. Specific scan parameters will not be made available to teams, only scan manufacturer.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

Data acquisition includes MRI data acquired on 1.5T and 3T scanners. For the primary modality (T1-weighted MRI), the vast majority will have 1 mm isotropic voxel sizes, but some sites will vary within standard parameter ranges. This year additional modalities will be made available as well for algorithm training (e.g. diffusion MRI, FLAIR, etc). The parameters for these will vary more.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

Baylor College of Medicine, University of California Los Angeles, Pennsylvania State University, Kennedy Krieger Institute, Loma Linda University, Nationwide Children's Hospital, Murdoch Children's Research Institute, Deakin University, University of Texas Houston, Kessler Foundation, VA Palo Alto, and the University of Oslo.

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

Image acquisition was performed by trained health professionals/radiologists at each respective institution.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

A case refers to one multimodal (mm) MRI scan from a particular patient. Some patients will have longitudinal data but "case" refers to individual mmMRI timepoint. The ground-truth (binary lesion masks) will be provided for the training cases only.

b) State the total number of training, validation and test cases.

The final AIMS-TBI 2025 dataset comprised a total of 892 images, including 553 images for training, 100 for validation, and 239 for testing. Data come from 13 different sites, with approximately the same distribution of training, validation, and test data randomly selected across sites. All cases in the test set are different from those in the training set.

c) How much of the data are already annotated (stratified by train test in percentage)?

All of the T1 data are already annotated for the training, validation, and testing data. The multimodal data will not have separate lesion masks, they will be made available only for the training data in an effort to allow algorithms to parse more detail from the T1w MRI and reduce noise.

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

553 training datasets - a large number of datasets to train a reliable model

100 validation datasets - a relatively small number of datasets to fine tune models and ensure fairness

239 test datasets - a relatively large number of datasets for an accurate, fair, final assessment of model performance

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The final test has been created to mirror a real-world distribution capturing variances in lesion heterogeneity, whilst also considering the balance of class distribution across training, validation, and testing environments. This strategy aims to create a realistic training and testing environment to develop algorithms capable of identifying heterogeneous lesion types present in real world ms-TBI research data sets.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

The data for the challenge will be the same as last year, with the exception of the additional modalities. The test dataset remains unseen.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

Manual annotation by 7 primary and 5 expert annotators. Primary annotators were staff at the University of Utah. Expert raters were Emily Dennis, Evelyn Deutscher, Karen Caeyenberghs, Hannah Lindsey, and Elisabeth Wilde.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

Training materials and instructions are summarized here:

https://drive.google.com/drive/folders/1LPoeCDBzCP_VfVOUoIPciCpHP8QdM1Gy

These are included as a link rather than written as they include a large number of visual aids.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

Staff at the University of Utah served as primary raters and were trained under the same protocol and completed segmentations on the same training data. Training data were reviewed by experts (ED and ED) and staff were approved to proceed to real data once they achieved a DSC of 0.6.

Lesion identification was performed in a 3 step process

- one primary rater reviewed the automated lesion segmentation result and performed manual edits
- a different primary rater reviewed the lesion segmented by the first rater and provided additional edits if necessary
- a third rater (expert rater) reviewed the lesion segmentation, provided minor edits if necessary, and approved the segmentations (any minor edits were communicated back to the primary raters to improve their following segmentations)

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

The annotations were created in a sequential process, after receiving the initial annotator's mask, the second annotator either approved it, or made edits, before sending it to the final expert who again either approved the mask, or made edits prior to approving it.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

The only pre-processing will be defacing to protect subject privacy. The goal is for the resulting algorithms to be able to handle data from various sources, so no other preprocessing has been done.

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

The primary source of error would be human error in image annotation. Statistical estimation of error is beyond the scope of this challenge, however, and will be evaluated in the future.

b) In an analogous manner, describe and quantify other relevant sources of error.

N/A

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)
- Absolute lesion count difference - Count (lesion-wise) = Total predicted - total rater
- Detection rate (lesion-wise DSC)
- Sensitivity and specificity
- Balanced accuracy (average of sensitivity and specificity)

Detection will be calculated for all lesions over 10 voxels. Lesions smaller than 10 voxels are below the reliable detection threshold given partial volume effects at standard acquisition resolutions and have poor inter-rater reliability.

The script that will be used for assessing algorithms will be made available to teams and all metrics will be detailed on the challenge website.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

The metrics chosen were selected to assess detection accuracy. Due to the large heterogeneity in patient presentations and injuries in ms-TBI, identifying when there is not a lesion present is also important, not only accurate segmentation of lesions that are present.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

We will leverage the same statistical framework that has been used by BraTS, which provides a fair, transparent, and robust evaluation framework building upon two stages: (i) a ranking score generated for each evaluated method, based on case-wise cumulative rankings aggregated across the pre-specified metrics reported above and across unseen hold-out cases, and (ii) a rigorous statistical significance assessment, based on pairwise permutation testing across the ranked order of the evaluated models.

b) Describe the method(s) used to manage submissions with missing results on test cases.

For instances where submitted algorithms fail to produce a mask/label for a specific test case, the metrics for that test case will be set to their worst possible value.

c) Justify why the described ranking scheme(s) was/were used.

In prior years the goal was to find the algorithm that performed best for both detection and segmentation, but following discussions with participants and within the organizing team, we have decided to separate detection and segmentation on the leaderboard this year. Reasons for this are - detection is more important for small lesions than segmentation, while segmentation becomes more critical with large lesions. There is enormous heterogeneity in lesions due to TBI, so splitting will allow us to optimize each task now. When the results from these tasks have improved we can return to a merged assessment.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

We will leverage the same statistical framework we have applied in prior years, which provides a fair, transparent, and robust evaluation framework building upon two stages: (i) a ranking score generated for each evaluated method, based on case-wise cumulative rankings aggregated across the pre-specified metrics reported above and across unseen hold-out cases, and (ii) a rigorous statistical significance assessment, based on pairwise permutation testing across the ranked order of the evaluated models.

Compared to a simplistic ordinal ranking, this two-staged methodology provides a more holistic assessment of model performance, identifying if performance differences between any two models are statistically significant and systematic, or the result of chance on a particular test set.

The article describing in detail the exact method has been accepted and will be released as part of the MICCAI BraTS 2025 Springer LNCS proceedings.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

Based on pairwise permutation testing across the ranked order of the evaluated models.

Provide a description of how variability of the performance of individual algorithms across test cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

SD and IQR across test cases – visualized as violin plots in our meta-analysis article and presented at the time of the challenge. Furthermore, we report the mean, standard errors and outliers.

Provide a description of how variability of rankings is assessed.

Based on pairwise permutation testing across the ranked order of the evaluated models.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

One-sided test assessment – the methodological details of the test for significance span over 2 pages of the accepted article. Springer informed us that this article should be released within January 2026.

To maintain statistical rigor, a correction for multiple comparisons, such as the Bonferroni correction or controlling the False Discovery Rate, will be applied to the p-values before interpretation.

Provide a description of the missing data handling.

N/A, as if an algorithm fails to produce a result metric for a specific test case, we set the scores to the worst possible rank for this case.

Indicate any software product that is used for all data analysis methods.

The software that we will use will be released as an open-source tool through GitHub together with the release of the BraTS article in January 2026.

We will provide pointers to the article and the software in the challenge website.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

N/A

TASK 2: Segmentation

SUMMARY

Abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Moderate to Severe Traumatic Brain Injury (msTBI) is caused by external forces (eg: traffic accidents, falls, sports) causing the brain to move rapidly within the skull, resulting in complex pathophysiological changes. Primary injuries arising from the initial forces (e.g., haematoma, hemorrhages, and contusions) (Mckee & Daneshvar, 2015), subsequently induce a cascade of secondary injuries (e.g., gliosis, encephalomalacia, (Maas et al., 2008)) that can include life threatening disorders such as raised intracranial pressure, requiring acute surgical intervention (Bullock et al., 2006). Each of these primary, secondary and surgery related processes can cause structural deformation in the brain. Each patient with msTBI has a unique accumulation of these structural changes, contributing to extremely heterogeneous lesions, considered a hallmark of msTBI (Covington & Duff, 2021). These lesions differ from other common brain pathologies (stroke, multiple sclerosis (MS), brain tumor) as they can be both focal or diffuse, varying in size, number and laterality, extending through multiple tissue types (e.g., grey matter (GM), white matter (WM), cerebrospinal fluid (CSF)), and can also occur in homologous regions of both hemispheres. Lesions such as these can complicate image registration, normalization, and are known to introduce both local and global errors in brain parcellation (Diamond et al., 2020; King et al., 2020). While several tools exist to compensate for lesions in neuroimaging pre-processing (i.e., the high definition brain extraction tool (HD_Bet) (Isensee et al., 2019), virtual brain grafting (VBG)(Radwan et al., 2021)), most require time consuming manual creation of lesion masks and subsequent manual quality assessment. Furthermore, in our experience, automated methods that have been developed for lesions of different etiologies (e.g. stroke, tumors [Sanjuán et al., 2013]) often underperform in TBI cases.

In the absence of appropriate processing tools, current lesion compensation techniques in msTBI include ignoring the lesions (resulting in unreliable findings), excluding msTBI patients with large lesions (limiting the generalizability), or manual segmentation of lesions prior to analysis (time consuming). Additionally, manual segmentation is often only feasible in smaller, single-site studies which lack the statistical power to perform subgroup analyses (i.e., divide the TBI sample according to lesion characteristics). These restrictive approaches limit the ability to investigate how factors such as type of injury (axonal injury, focal lesions, and diffuse microlesions), severity (mild, moderate, severe), and presence of comorbid injuries or complications (especially those that affect pulmonary and cardiovascular function, see (Crawford et al., 2019) and post-traumatic seizure [Bennett et al., 2017; Liesemer et al., 2011]) impact patient's functional outcomes. To capture this information, msTBI researchers need access to an accurate, automatic lesion segmentation algorithm trained on large multicohort MRI datasets, where images are aggregated across sites. Whilst a handful of TBI specific algorithms exist, they require either multiple image types (Kamnitsas et al., 2017) (T1, T2, FLAIR, GE & PD) or can run on only CT images (Jain et al., 2019). The necessity for multiple image types limits the applicability of such tools to large-scale consortia aggregating common MRI scans across sites.

The AIMS-TBI challenge leverages data shared within the Enhancing NeuroImaging Genetics through

Meta-Analysis (ENIGMA) Consortium (Thompson et al., 2022) Traumatic Brain Injury working group, specifically, the Pediatric mTBI and Adult mTBI subgroups. Therefore, the AIMS-TBI challenge will focus on identifying lesions in T1-weighted (T1w) MRI data only as it is both the most common MRI scan across our ENIGMA TBI consortium, and exhibits less parameter variation (e.g. 1 mm³ voxel size was relatively common in our previous published work, Dennis et al., 2023; Keleher et al., 2022) compared to other MRI modalities (e.g., diffusion MRI) which therefore results in more comparable data collected across sites.

The inaugural AIMS-TBI Challenge was held at the 2024 MICCAI conference and it was held again in 2025. The final AIMS-TBI 2025 dataset comprised a total of 892 images, including 553 images for training, 100 for validation, and 239 for testing. In the validation phase of the challenge, 15 teams uploaded 60 entries, with 9 teams submitting final models in the test stage. The best performing model achieved an average Dice score of 0.637 overall, and the highest Dice for the lesion-only files being 0.515. This result represented an improvement over the first year, yet also highlighted substantial room for continued development. This year we will separate the tasks of segmentation and detection, identifying which subjects are lesion-free in the training data and focusing the performance metrics on the cases with lesions and having dual leaderboards for each task. Additionally, this year we will make multimodal MRI data available for teams to use in training their algorithms.

Keywords

List the primary keywords that characterize the task.

T1-weighted MRI, TBI, traumatic brain injury, lesion, segmentation

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Emily Dennis - University of Utah

Evelyn Deutscher - Deakin University

Elisabeth Wilde - University of Utah

Matthew Pease - Indiana University

Ujjwal Baid - Emory University

Spyridon Bakas - Indiana University

Adrian Onicas - University of Utah

b) Provide information on the primary contact person.

Emily Dennis, emily.dennis@hsc.utah.edu

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Yes Matthew Pease is a neurosurgeon with experience in surgical interventions for TBI. He assisted with designing the challenge in 2025 and gave a lecture during the challenge on lesions in TBI from a clinical perspective. He continues to play a central role in designing the challenge to be clinically relevant and biologically based, and assists with the review of the reference standard and interpretation of results.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place. Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

Repeated event with annual fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

MICCAI 2026

b) Report the platform used to run the challenge.

Synapse

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

New challenge website has not been established yet. We are shifting from Grand Challenge (which we used in 2024 and 2025) to Synapse as the challenge is run in closer coordination with BraTS

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

Users are allowed to conduct preprocessing steps if they are done with a readily available tool.

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

Publicly available data is allowed.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard

d) Define the award policy. In particular, provide details with respect to challenge prizes.

The top 3 performing teams will receive a cash prize - \$500 USD for first place, \$325 for second place, \$175 for third place.

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

Algorithm performance will be posted on the challenge website during the validation phase and updated with each submission. Final performance will be announced at the MICCAI in-person challenge event and the top 3 teams will be listed on the challenge website at this time.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

Participating teams will be asked to submit a 8-12 page paper on their algorithms and results after the validation phase, due July 31st, as part of the challenge. The top three teams will be invited to work with the challenge organizers on a challenge paper to be submitted to a medical imaging journal after the conclusion of the challenge.

Teams may publish any additional papers after submission of the challenge paper.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Algorithm container on the Synapse platform. <https://www.synapse.org/aimstbi2026> - (Website will be publicly visible after the challenge approval).

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

We plan to release a validation subset 6 weeks after the release of the training data. The ground truth data for the validation set will not be released, but 3 submissions will be allowed for groups to fine tune their algorithms prior to the final test.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

16 March 2026 (or earlier): Registration opens

Release of Training data

Participants will be able to register for the challenge in synapse.org, from the date of its potential acceptance (either Feb 9, or Mar 16, 2026) until the short paper submission deadline (July 31, 2026).

30 April 2026: Release of Validation data

31 July 2026: Submission deadline for Paper and Containerized method

Paper should include motivation, literature, method description, and results on training and validation data. The containerized method will be evaluated on the hidden testing data by the organizers, only for participants with submitted short papers. Ranking of all participating methods, following statistical significance assessment based on multiple permutation testing.

7 August 2026: Invitations are sent

Inviting participants with a valid submitted paper and an evaluated container to present at the conference. Teams will be notified at this point of their rank and the top 3 teams will be invited to give an oral presentation.

14 August 2026: Paper reviews are available

19 August 2026: Release of summary testing results to each participant.

Participants will receive summary testing data results, in order to report those in their papers.

21 August 2026: Camera-ready paper and Copyright form submission deadline

1 September 2026: Organizers to send all proceedings material to the MICCAI 2026 Satellite Event committee for inclusion in the MICCAI LNCS Springer Proceedings.

4-8 October 2026: Challenge at MICCAI

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

University of Utah IRB has acknowledged the ENIGMA projects as not human subjects due to data de-identification. The University of Utah has DUAs/MTAs/CRADAs/SCCs in place with each site that has contributed data. Data will be further anonymized with the following steps:

- MRI data are defaced using pydeface with any subject or site data stripped from the metadata
- New ID codes were created, with original subject IDs recoded with the key securely held at the University of Utah
- Data is pooled across sites with no way to re-identify site
- The only demographic data provided along with the MRI data are age (rounded to the nearest year), sex, and time since injury (in weeks)

With these steps, we are confident that subject/site re-identification will be impossible. Participating challenge teams will need to complete a MOU with the University of Utah prior to receiving access to the data.

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)
- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

A previous script for evaluating results can be found here: <https://github.com/adionicas/AIMS-TBI-challenge>. A more formal, updated script will be released through our Synapse website.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

The participants are required to submit their containerized algorithm, during or after the validation phase. Specific instructions for the containerization will be provided after the challenge approval. All challenge participants will be required to accept an agreement through the synapse.org website that participation in the testing phase of the challenge, will automatically mean that we can make their containerized method publicly available through our challenge webpage and a collective toolkit we are putting together.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

The National Institutes of Health funds this work. Emily Dennis, Evelyn Deutscher, and Adrian Onicas will have access to the test case labels.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Research,Treatment planning,Diagnosis,Assistance,Surgery,Intervention planning,Education,Training

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation

- Tracking

Segmentation

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort would be patients who have sustained a moderate-severe TBI who have MRI data.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The challenge cohort includes subjects from the ENIGMA Pediatric mTBI and ENIGMA Adult mTBI working groups who have MRI data that has been shared with the University of Utah.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

T1-weighted brain MRI, diffusion MRI, FLAIR, SWI, T2

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

Scan manufacturer and segmentation mask.

b) ... to the patient in general (e.g. sex, medical history).

Patient age, sex, time since injury

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

Brain MRI scan

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

3D lesions in the brain

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Find highly accurate lesion segmentation algorithm for lesions due to moderate/severe TBI

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

The MRI data used in this challenge come from multiple MRI scanner manufacturers (GE, Siemens, Philips), multiple scan models, and differing sets of parameters. Specific scan parameters will not be made available to teams, only scan manufacturer.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

Data acquisition includes MRI data acquired on 1.5T and 3T scanners. For the primary modality (T1-weighted MRI), the vast majority will have 1 mm isotropic voxel sizes, but some sites will vary within standard parameter ranges. This year additional modalities will be made available as well for algorithm training (e.g. diffusion MRI, FLAIR, etc). The parameters for these will vary more.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

Baylor College of Medicine, University of California Los Angeles, Pennsylvania State University, Kennedy Krieger Institute, Loma Linda University, Nationwide Children's Hospital, Murdoch Children's Research Institute, Deakin University, University of Texas Houston, Kessler Foundation, VA Palo Alto, and the University of Oslo.

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

Image acquisition was performed by trained health professionals/radiologists at each respective institution.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

A case refers to one multimodal (mm) MRI scan from a particular patient. Some patients will have longitudinal data but "case" refers to individual mmMRI timepoint. The ground-truth (binary lesion masks) will be provided for the training cases only.

b) State the total number of training, validation and test cases.

The final AIMS-TBI 2025 dataset comprised a total of 892 images, including 553 images for training, 100 for validation, and 239 for testing. Data come from 13 different sites, with approximately the same distribution of training, validation, and test data randomly selected across sites. All cases in the test set are different from those in the training set.

c) How much of the data are already annotated (stratified by train test in percentage)?

All of the T1 data are already annotated for the training, validation, and testing data. The multimodal data will not have separate lesion masks, they will be made available only for the training data in an effort to allow algorithms to parse more detail from the T1w MRI and reduce noise.

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

553 training datasets - a large number of datasets to train a reliable model

100 validation datasets - a relatively small number of datasets to fine tune models and ensure fairness

239 test datasets - a relatively large number of datasets for an accurate, fair, final assessment of model performance

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The final test has been created to mirror a real-world distribution capturing variances in lesion heterogeneity, whilst also considering the balance of class distribution across training, validation, and testing environments. This strategy aims to create a realistic training and testing environment to develop algorithms capable of identifying heterogeneous lesion types present in real world ms-TBI research data sets.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

The data for the challenge will be the same as last year, with the exception of the additional modalities. The test dataset remains unseen.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

Manual annotation by 7 primary and 5 expert annotators. Primary annotators were staff at the University of Utah. Expert raters were Emily Dennis, Evelyn Deutscher, Karen Caeyenberghs, Hannah Lindsey, and Elisabeth Wilde.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

Training materials and instructions are summarized here:

https://drive.google.com/drive/folders/1LPoeCDBzCP_VfVOUoIPciCpHP8QdM1Gy

These are included as a link rather than written as they include a large number of visual aids.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

Staff at the University of Utah served as primary raters and were trained under the same protocol and completed segmentations on the same training data. Training data were reviewed by experts (ED and ED) and staff were approved to proceed to real data once they achieved a DSC of 0.6.

Lesion identification was performed in a 3 step process

- one primary rater reviewed the automated lesion segmentation result and performed manual edits
- a different primary rater reviewed the lesion segmented by the first rater and provided additional edits if necessary
- a third rater (expert rater) reviewed the lesion segmentation, provided minor edits if necessary, and approved the segmentations (any minor edits were communicated back to the primary raters to improve their following segmentations)

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

The annotations were created in a sequential process, after receiving the initial annotator's mask, the second annotator either approved it, or made edits, before sending it to the final expert who again either approved the mask, or made edits prior to approving it.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

The only pre-processing will be defacing to protect subject privacy. The goal is for the resulting algorithms to be able to handle data from various sources, so no other preprocessing has been done.

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

The primary source of error would be human error in image annotation. Statistical estimation of error is beyond the scope of this challenge, however, and will be evaluated in the future.

b) In an analogous manner, describe and quantify other relevant sources of error.

N/A

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

- Dice similarity coefficient - DSC (voxelwise) = $2TP / (2TP + FN + FP)$

Only for lesion-containing datasets, only calculated on lesions over XX voxels

- Hausdorff distance 95%ile (HD95)

- Average symmetric surface distance (ASSD)

Segmentation will be calculated for all lesions over 50 voxels. A larger threshold was chosen for segmentation because accuracy is disproportionately affected for small lesions. A histogram of lesion sizes showed a heavily left-skewed distribution, as expected, but showed a reasonable cut-point of 50 voxels. There are numerous small but significant structures ~50 voxels, such as some thalamic nuclei and the locus coeruleus.

The script that will be used for assessing algorithms will be made available to teams and all metrics will be detailed on the challenge website.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

The metrics chosen were selected to detect segmentation accuracy. DSC to measure overlap, HD95 and ASSD as boundary-based distance algorithms. HD95 and ASSD were selected to measure both average and maximum boundary error.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

We will leverage the same statistical framework that has been used by BraTS, which provides a fair, transparent, and robust evaluation framework building upon two stages: (i) a ranking score generated for each evaluated method, based on case-wise cumulative rankings aggregated across the pre-specified metrics reported above and across unseen hold-out cases, and (ii) a rigorous statistical significance assessment, based on pairwise permutation testing across the ranked order of the evaluated models.

b) Describe the method(s) used to manage submissions with missing results on test cases.

For instances where submitted algorithms fail to produce a mask/label for a specific test case, the metrics for that test case will be set to their worst possible value (e.g., 0 for the DSC).

c) Justify why the described ranking scheme(s) was/were used.

In prior years the goal was to find the algorithm that performed best for both detection and segmentation, but following discussions with participants and within the organizing team, we have decided to separate detection and segmentation on the leaderboard this year. Reasons for this are - detection is more important for small lesions than segmentation, while segmentation becomes more critical with large lesions. There is enormous heterogeneity in lesions due to TBI, so splitting will allow us to optimize each task now. When the results from these tasks have improved we can return to a merged assessment.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

We will leverage the same statistical framework we have applied in prior years, which provides a fair, transparent, and robust evaluation framework building upon two stages: (i) a ranking score generated for each evaluated method, based on case-wise cumulative rankings aggregated across the pre-specified metrics reported above and across unseen hold-out cases, and (ii) a rigorous statistical significance assessment, based on pairwise permutation testing across the ranked order of the evaluated models.

Compared to a simplistic ordinal ranking, this two-staged methodology provides a more holistic assessment of model performance, identifying if performance differences between any two models are statistically significant and systematic, or the result of chance on a particular test set.

The article describing in detail the exact method has been accepted and will be released as part of the MICCAI BraTS 2025 Springer LNCS proceedings.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

Based on pairwise permutation testing across the ranked order of the evaluated models.

Provide a description of how variability of the performance of individual algorithms across test cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

SD and IQR across test cases – visualized as violin plots in our meta-analysis article and presented at the time of the challenge. Furthermore, we report the mean, standard errors and outliers.

Provide a description of how variability of rankings is assessed.

Based on pairwise permutation testing across the ranked order of the evaluated models.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

One-sided test assessment – the methodological details of the test for significance span over 2 pages of the accepted article. Springer informed us that this article should be released within January 2026.

To maintain statistical rigor, a correction for multiple comparisons, such as the Bonferroni correction or

controlling the False Discovery Rate, will be applied to the p-values before interpretation.

Provide a description of the missing data handling.

N/A, as if an algorithm fails to produce a result metric for a specific test case, we set the scores to the worst possible rank for this case.

Indicate any software product that is used for all data analysis methods.

The software that we will use will be released as an open-source tool through GitHub together with the release of the BraTS article in January 2026.

We will provide pointers to the article and the software in the challenge website.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

N/A

ADDITIONAL POINTS

References

Please include any reference important for the challenge design, for example publications on the data, the annotation process or the chosen metrics as well as DOIs referring to data or code.

N/A

Further comments

Further comments from the organizers.

N/A